Cloud Infrastructure & Architecture Blueprint

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Module 1: Microservices Architecture and Service Mesh

The backend architecture comprises decoupled containerized microservices hosted on multi-tenant Kubernetes clusters. Ingress routing and traffic routing are managed through an Envoy-based API Gateway enforcing strict rate limiting of 10,000 requests per second per tenant.

Service-to-service communication is orchestrated via an Istio service mesh, providing mutual TLS (mTLS) encryption by default. All ingress API endpoints require OAuth 2.0 JWT bearer tokens validated against our centralized identity provider.

Module 2: Vector Database and Semantic Search Tier

A vector database is an optimized storage and retrieval engine that indexes high-dimensional vector embeddings generated by machine learning models. Unlike traditional relational databases that match exact keywords or scalar attributes, a vector database performs approximate nearest neighbor (ANN) search to discover semantically similar passages based on cosine distance or dot-product similarity.

Our RAG search tier leverages HNSW (Hierarchical Navigable Small World) indexing to maintain sub-50 millisecond retrieval latency across millions of chunk vectors. A distributed Redis cache layer acts as a semantic cache with a cosine similarity threshold of 0.92, bypassing vector index computation for repeated queries.

Module 3: High Availability and Disaster Recovery (DR)

To ensure business continuity, our cloud services run in an active-active multi-region configuration across US-East and US-West regions. Database replication utilizes synchronous multi-AZ replicas with asynchronous cross-region streaming.

Our Disaster Recovery Service Level Agreements (SLAs) enforce a Recovery Point Objective (RPO) of under 5 minutes and a Recovery Time Objective (RTO) of under 15 minutes. Automated failover health checks trigger DNS reconfiguration via Route53 when 3 consecutive probes fail.